Skewness and Kurtosis

Measuring the asymmetry and tail behaviour of a distribution with quantiles and moments.

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1 Skewness: the idea

Skewness is the degree of asymmetry of a distribution.

Shape	Tail	Typical order*
Symmetric	Balanced	Mean = Median = Mode
Positive	Longer **right** tail	Mean > Median > Mode
Negative	Longer **left** tail	Mean < Median < Mode

*For many unimodal distributions; not a universal rule.

2 Karl Pearson's coefficient

Using the mode

$$S_k = \frac{\text{Mean} - \text{Mode}}{\text{SD}}$$

Using the median

$$S_k = \frac{3(\text{Mean} - \text{Median})}{\text{SD}}$$

Based on the mean, the mode (or median) and the standard deviation. Sign gives the direction: $+$ positive, $-$ negative, 0 symmetric.

3 Bowley's (quartile) measure

Based on quartiles

$$S_Q = \frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1}$$

Equivalent: $\frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 - Q_2) + (Q_2 - Q_1)}$. Lies between −1 and +1; usable with open-end classes.

4 Kelley's (percentile) measure

Based on P_{10}, P_{50}, P_{90}

$$S_P = \frac{(P_{90} - P_{50}) - (P_{50} - P_{10})}{(P_{90} - P_{50}) + (P_{50} - P_{10})}$$

Uses more of the distribution than Bowley's middle 50%.

5 Central moments

rth central moment

$$\mu_r = \frac{\sum f_i (x_i - \bar{x})^r}{N}$$

μ_0	μ_1	μ_2	μ_3	μ_4
1	0 (always)	Variance σ^2	Skewness	Kurtosis

6 Moment skewness

Size only

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

Size and direction

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}}$$

$\gamma_1 = 0$ means zero third-moment skewness; it does not prove symmetry.

7 Kurtosis

Based on the **fourth** standardised central moment; describes tail behaviour and extreme observations.

Kurtosis

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

Excess kurtosis

$$\gamma_2 = \beta_2 - 3$$

Type	β_2	γ_2	Example
Mesokurtic	= 3	= 0	Normal distribution
Leptokurtic	> 3	> 0	Heavier tails, sharper peak
Platykurtic	< 3	< 0	Lighter tails, flatter

! Common mistakes

- × β_1 has no sign; use γ_1 for direction.
- × The skewed mean–median–mode order is typical, not guaranteed.
- × "Positive skew" means the long tail is on the **right**.

? Fill in the blanks

1. Mean 52, mode 46 and SD 12 give Karl Pearson's coefficient S_k = _____. [0.5]
2. Mean 30, median 33 and SD 6 give S_k = _____, so the distribution is _____ skewed. [−1.5; negatively]
3. If Q_1 = 10, Q_2 = 15 and Q_3 = 20, Bowley's coefficient of skewness is _____. [0]
4. A distribution with β_2 = 4.2 is called _____. [leptokurtic]
5. If μ_2 = 4 and μ_4 = 48, then β_2 = _____, so the curve is mesokurtic. [3]
6. β_1 cannot show the direction of skewness because it is never _____. [negative]

Source: *Textbook of Agricultural Statistics*, Ch. 7 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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